

WEB PROGRAMMING LAB-10&11

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Title: Lab 10: Databases - Part I

Title: Database Integration and Model Creation in Django

Question 1

Design a web site using Django, which is a website directory - A site containing links to other websites. A web page has different categories. A category table has a name, number of visits, and number of likes. A page table refers to a category, has a title, URL, and many views. Design a form that populates the above database and displays it.

Create project & App

Create Django project

```
django-admin startproject directoryproject
cd directoryproject
```

Create app

```
python manage.py startapp directory
```

Project Settings: `directoryproject/settings.py`

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'directory',
]
```

Database Initialization

```
python manage.py migrate
```

Project URLs: `directoryproject/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
```

```

    path('admin/', admin.site.urls),
    path('', include('directory.urls')),
]

```

App URLs: `directory/urls.py`

```

from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
    path('add_category/', views.add_category, name='add_category'),
    path('add_page/', views.add_page, name='add_page'),
]

```

Models: `directory/models.py`

```

from django.db import models

class Category(models.Model):
    name = models.CharField(max_length=128)
    visits = models.IntegerField(default=0)
    likes = models.IntegerField(default=0)

    def __str__(self):
        return self.name

class Page(models.Model):
    category = models.ForeignKey(Category, on_delete=models.CASCADE)
    title = models.CharField(max_length=128)
    url = models.URLField()
    views = models.IntegerField(default=0)

    def __str__(self):
        return self.title

```

Forms: `directory/forms.py`

```

from django import forms
from .models import Category, Page

class CategoryForm(forms.ModelForm):
    class Meta:
        model = Category
        fields = ['name', 'visits', 'likes']

class PageForm(forms.ModelForm):
    class Meta:
        model = Page
        fields = ['category', 'title', 'url', 'views']

```

Views: `directory/views.py`

```

from django.shortcuts import render, redirect
from .models import Category, Page
from .forms import CategoryForm, PageForm

def home(request):
    categories = Category.objects.all()
    pages = Page.objects.all()
    return render(request, 'home.html', {
        'categories': categories,
        'pages': pages
    })

def add_category(request):
    if request.method == 'POST':
        form = CategoryForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')
    else:
        form = CategoryForm()
    return render(request, 'add_category.html', {'form': form})

def add_page(request):
    if request.method == 'POST':
        form = PageForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')
    else:
        form = PageForm()
    return render(request, 'add_page.html', {'form': form})

```

Templates

directory/templates/home.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Website Directory</title>
</head>
<body>
    <h1>Website Directory</h1>
    <p>
        <a href="/add-category/">Add Category</a> |
        <a href="/add-page/">Add Page</a>
    </p>
    <hr>

    {% for category in categories %}
        <h2>{{ category.name }}</h2>
        <p>Visits: {{ category.visits }} | Likes: {{ category.likes }}</p>
        <ul>
            {% for page in category.page_set.all %}

```

```

        <li>
            <strong>{{ page.title }}</strong><br>
            URL: <a href="{{ page.url }}" target="_blank">{{ page.url }}</a><b
r>
            Views: {{ page.views }}
        </li>
    {% empty %}
        <li>No pages in this category.</li>
    {% endfor %}
</ul>
<hr>
{% endfor %}
</body>
</html>

```

directory/templates/add_category.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Add Category</title>
</head>
<body>
    <h1>Add Category</h1>
    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Save</button>
    </form>
    <br>
    <a href="/">Back to Home</a>
</body>
</html>

```

directory/templates/add_page.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Add Page</title>
</head>
<body>
    <h1>Add Page</h1>
    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Save</button>
    </form>
    <br>
    <a href="/">Back to Home</a>
</body>
</html>

```

Run

```
python manage.py runserver 8080
```

Question 2

Consider the following tables: WORKS(person-name, Company-name, Salary) LIVES(Person_name, Street, City)
Assume Table data suitably. Design a Django webpage and include an option to insert data into WORKS table by accepting data from the user using TextBoxes. Also, include an option to retrieve the names of people who work for a particular company along with the cities they live in (particular company name must be accepted from the user).

Create project & App

Create Django project

```
django-admin startproject companyproject  
cd companyproject
```

Create app

```
python manage.py startapp companyapp
```

Project Settings: `companyproject/settings.py`

```
INSTALLED_APPS = [  
    'companyapp',  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
]
```

Database Initialization

```
python manage.py migrate
```

Project URLs: `companyproject/urls.py`

```
from django.contrib import admin  
from django.urls import path, include  
  
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('', include('companyapp.urls')),  
]
```

App URLs: `companyapp/urls.py`

```
from django.urls import path  
from . import views  
  
urlpatterns = [  
    path('add/', views.add_person),  
    path('get/', views.get_persons),  
]
```

```

    path('', views.home, name='home'),
    path('add/', views.add_work, name='add_work'),
    path('search/', views.search_company, name='search_company'),
]

```

Models: `companyapp/models.py`

```

from django.db import models

# WORKS table
class Works(models.Model):
    person_name = models.CharField(max_length=100)
    company_name = models.CharField(max_length=100)
    salary = models.IntegerField()

    def __str__(self):
        return self.person_name

# LIVES table
class Lives(models.Model):
    person_name = models.CharField(max_length=100)
    street = models.CharField(max_length=100)
    city = models.CharField(max_length=100)

    def __str__(self):
        return self.person_name

```

Admin: `companyapp/admin.py`

```

from django.contrib import admin
from .models import Works, Lives

admin.site.register(Works)
admin.site.register(Lives)

```

Forms: `companyapp/forms.py`

```

from django import forms
from .models import Works

# Form for inserting WORKS data
class WorksForm(forms.ModelForm):
    class Meta:
        model = Works
        fields = ['person_name', 'company_name', 'salary']

# Form for searching company
class SearchForm(forms.Form):
    company_name = forms.CharField(max_length=100)

```

Views: `companyapp/views.py`

```

from django.shortcuts import render, redirect
from .models import Works, Lives
from .forms import WorksForm, SearchForm

# Insert into WORKS table
def add_work(request):
    form = WorksForm()

    if request.method == 'POST':
        form = WorksForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')

    return render(request, 'add_work.html', {'form': form})

# Search employees by company
def search_company(request):
    results = []
    form = SearchForm()

    if request.method == 'POST':
        form = SearchForm(request.POST)
        if form.is_valid():
            company = form.cleaned_data['company_name']
            works = Works.objects.filter(company_name=company)

            for w in works:
                try:
                    city = Lives.objects.get(person_name=w.person_name).city
                except:
                    city = "Not Found"

                results.append({
                    'name': w.person_name,
                    'city': city
                })

    return render(request, 'search.html', {
        'form': form,
        'results': results
    })

# Home page
def home(request):
    return render(request, 'home.html')

```

Templates

companyapp/templates/home.html

```

<html>
<body>

```

```

<h2>Company Database</h2>

<a href="/add/">Add Employee Work</a><br><br>
<a href="/search/">Search by Company</a>

</body>
</html>

```

companyapp/templates/add_work.html

```

<html>
<body>

<h2>Add WORKS Data</h2>

<form method="POST">
{% csrf_token %}
{{ form.as_p }}
<button type="submit">Save</button>
</form>

</body>
</html>

```

companyapp/templates/search.html

```

<html>
<body>

<h2>Search Employees by Company</h2>

<form method="POST">
{% csrf_token %}
{{ form.as_p }}
<button type="submit">Search</button>
</form>

<br>

{% if results %}
<h3>Results:</h3>
<ul>
{% for r in results %}
<li>{{ r.name }} - {{ r.city }}</li>
{% endfor %}
</ul>
{% endif %}

</body>
</html>

```

Run

```
python manage.py runserver 8080
```

Lab 11: Databases - Part II

Question 1

There are three tables in the database an author table has a first name, a last name and an email address. A publisher table has a name, a street address, a city, a state/ province, a country, and a Web site. A book table has a title and a publication date. It also has one or more authors (a many-to-many relationship with authors) and a single publisher (a one-to-many relationship - aka foreign key - to publishers). Design a form which populates and retrieves the information from the above database using Django.

Create project & App

Create Django project

```
django-admin startproject LibraryProject
cd LibraryProject
```

Create app

```
python manage.py startapp library
```

Project Settings: **LibraryProject/settings.py**

```
INSTALLED_APPS = [
    'library',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]
```

Database Initialization

```
python manage.py makemigrations
python manage.py migrate
```

Project URLs: **LibraryProject/urls.py**

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('library.urls')),
]
```

App URLs: `library/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
    path('author/', views.add_author, name='add_author'),
    path('publisher/', views.add_publisher, name='add_publisher'),
    path('book/', views.add_book, name='add_book'),
    path('show/', views.show_data, name='show'),
]
```

Models: `library/models.py`

```
from django.db import models

# Author Table
class Author(models.Model):
    first_name = models.CharField(max_length=100)
    last_name = models.CharField(max_length=100)
    email = models.EmailField()

    def __str__(self):
        return self.first_name

# Publisher Table
class Publisher(models.Model):
    name = models.CharField(max_length=200)
    street_address = models.CharField(max_length=200)
    city = models.CharField(max_length=100)
    state = models.CharField(max_length=100)
    country = models.CharField(max_length=100)
    website = models.URLField()

    def __str__(self):
        return self.name

# Book Table
class Book(models.Model):
    title = models.CharField(max_length=200)
    publication_date = models.DateField()

    authors = models.ManyToManyField(Author)
    publisher = models.ForeignKey(Publisher, on_delete=models.CASCADE)

    def __str__(self):
        return self.title
```

Admin: `library/admin.py`

```
from django.contrib import admin
from .models import Author, Publisher, Book
```

```
admin.site.register(Author)
admin.site.register(Publisher)
admin.site.register(Book)
```

Forms: `library/forms.py`

```
from django import forms
from .models import Author, Publisher, Book

class AuthorForm(forms.ModelForm):
    class Meta:
        model = Author
        fields = '__all__'

class PublisherForm(forms.ModelForm):
    class Meta:
        model = Publisher
        fields = '__all__'

class BookForm(forms.ModelForm):
    class Meta:
        model = Book
        fields = '__all__'
```

Views: `library/views.py`

```
from django.shortcuts import render, redirect
from .models import Author, Publisher, Book
from .forms import AuthorForm, PublisherForm, BookForm

def home(request):
    return render(request, 'home.html')

# Add Author
def add_author(request):
    form = AuthorForm()
    if request.method == 'POST':
        form = AuthorForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')
    return render(request, 'add_author.html', {'form': form})

# Add Publisher
def add_publisher(request):
    form = PublisherForm()
    if request.method == 'POST':
        form = PublisherForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')
    return render(request, 'add_publisher.html', {'form': form})

# Add Book
```

```
def add_book(request):
    form = BookForm()
    if request.method == 'POST':
        form = BookForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')
    return render(request, 'add_book.html', {'form': form})

# Display Data
def show_data(request):
    books = Book.objects.all()
    return render(request, 'show.html', {'books': books})
```

Templates

Library/templates/home.html

```
<html>
<body>

<h2>Library System</h2>

<a href="/author/">Add Author</a><br>
<a href="/publisher/">Add Publisher</a><br>
<a href="/book/">Add Book</a><br>
<a href="/show/">Show Books</a>

</body>
</html>
```

Library/templates/add_author.html

```
<html>
<body>

<h2>Add Author</h2>

<form method="POST">
{% csrf_token %}
{{ form.as_p }}
<button type="submit">Save</button>
</form>

</body>
</html>
```

Library/templates/add_publisher.html

```
<html>
<body>

<h2>Add Publisher</h2>
```

```

<form method="POST">
{% csrf_token %}
{{ form.as_p }}
<button type="submit">Save</button>
</form>

</body>
</html>

```

Library/templates/add_book.html

```

<html>
<body>

<h2>Add Book</h2>

<form method="POST">
{% csrf_token %}
{{ form.as_p }}
<button type="submit">Save</button>
</form>

</body>
</html>

```

Library/templates/show.html

```

<html>
<body>

<h2>Books List</h2>

<ul>
{% for book in books %}
    <li>
        {{ book.title }} - {{ book.publisher.name }} -
        {% for author in book.authors.all %}
            {{ author.first_name }}
        {% endfor %}
    </li>
{% endfor %}
</ul>

</body>
</html>

```

Run

```
python manage.py runserver 8080
```

Lab 11: Databases - Part II

Question 2

Create a Django Page for entry of a Product information (title, price and description) and save it into the db.
Create the index page where you would view the product entries in an unordered list.

Create project & App

Create Django project

```
django-admin startproject ProductProject
cd ProductProject
```

Create app

```
python manage.py startapp productapp
```

Project Settings: `ProductProject/settings.py`

```
INSTALLED_APPS = [
    'productapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]
```

Database Initialization

```
python manage.py makemigrations
python manage.py migrate
```

Project URLs: `ProductProject/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('productapp.urls')),
]
```

App URLs: `productapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index, name='index'),
    path('add/', views.add_product, name='add_product'),
]
```

Models: `productapp/models.py`

```

from django.db import models

class Product(models.Model):
    title = models.CharField(max_length=100)
    price = models.IntegerField()
    description = models.TextField()

    def __str__(self):
        return self.title

```

Admin: `productapp/admin.py`

```

from django.contrib import admin
from .models import Product

admin.site.register(Product)

```

Forms: `productapp/forms.py`

```

from django import forms
from .models import Product

class ProductForm(forms.ModelForm):
    class Meta:
        model = Product
        fields = ['title', 'price', 'description']

```

Views: `productapp/views.py`

```

from django.shortcuts import render, redirect
from .models import Product
from .forms import ProductForm

# Add Product
def add_product(request):
    form = ProductForm()

    if request.method == 'POST':
        form = ProductForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('index')

    return render(request, 'add_product.html', {'form': form})

# Index Page (Display Products)
def index(request):
    products = Product.objects.all()
    return render(request, 'index.html', {'products': products})

```

Templates

`productapp/templates/add_product.html`

```
<html>
<body>

<h2>Add Product</h2>

<form method="POST">
{% csrf_token %}
{{ form.as_p }}
<button type="submit">Save</button>
</form>

</body>
</html>
```

productapp/templates/index.html

```
<html>
<body>

<h2>Product List</h2>

<ul>
{% for product in products %}
    <li>
        {{ product.title }} - ₹{{ product.price }} - {{ product.description }}
    </li>
{% endfor %}
</ul>

<br>

<a href="/add/">Add New Product</a>

</body>
</html>
```

Run

```
python manage.py runserver 8080
```